

Compound interest tables



1

a

	Balance + interest	Total balance	Interest earned
First year	—	\$4400	\$220
Second year	$\$4400 + \220	\$4620	\$231
Third year	$\$4620 + \231	\$4851	\$242.55
Fourth year	$\$4851 + \242.55	\$5093.55	—

b \$693.55

2

a

	Balance + interest	Total balance	Interest earned
First year	—	\$3600	\$252
Second year	$\$3600 + \252	\$3852	\$269.64
Third year	$\$3852 + \269.64	\$4121.64	\$288.51
Fourth year	$\$4121.64 + \288.51	\$4410.15	—

b \$810.15

3

a

	Balance + interest	Total balance	Interest earned
First year	—	\$570	\$51.30
Second year	$\$570 + \51.30	\$621.30	\$55.92
Third year	$\$621.30 + \55.92	\$677.22	—

b \$107.22

4

a

	Balance + interest	Total balance	Interest earned
First year	—	\$6100	\$366
Second year	$\$6100 + \366	\$6466	\$387.96
Third year	$\$6466 + \387.96	\$6853.96	—

b \$753.96

5



a

Time Period n (years)	Value at beginning of time period	Value at end of time period	Interest earned in time period
1	\$1000	\$1020	\$20
2	\$1020	\$1040.40	\$20.40
3	\$1040.40	\$1061.21	\$20.81

b \$61.21

Compound interest

6

a True

b False

c True

d True

7

a \$81 033.75

b \$11 033.75

8

a \$647.20

b \$8737.20

c \$698.98

d \$9436.18

e \$754.89

f \$10 191.07

g \$2101.07

h \$25.97%

i \$1941.60

j \$159.47

9

a \$5894.42

b \$894.42

10 \$39 222.92

11 \$49 909.03

12

a $A(t) = 10\,000(1.013)^t$

b \$1009.40

Other compound periods

13

a \$8912.51

b \$5609

c \$1997.90

d \$5400.97

14 a \$2567.12 b \$559.56  c \$1400.60 d \$4096.44

15 a \$7761.19 b \$7714.56 c \$7690.89 d \$7674.97
e \$7668.82 f \$7667.23

16 a
i \$2265.59 ii \$2268.86 iii \$2270.13
b Compounding monthly earns \$3.27 more than compounding quarterly and compounding weekly earns \$1.27 more than compounding monthly.
c More interest is earned with more frequent compounding, but the extra interest earned decreases as the compounding frequency increases.

17 Plan 2 is the best option. We can either find the future value of each investment or calculate the effective annual interest rate of each plan.
Plan 1 has a future value of \$6625.24, Plan 2 has a future value of \$7050.97, and Plan 3 has a future value of \$6855.10.